

# Effect of Cordyceps Sinensis on Uncoupling Protein 2 in Experimental Rats with Nonalcoholic Fatty Liver

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[Objective] To investigate the effect of cordyceps sinensis (CS) on expression of uncoupling protein-2 (UCP2) and so to elucidate the role of UCP2 in development of nonalcoholic fatty liver diseases (NAFLD). [Methods] Rats were administrated with high-fat diet to produce NAFLD animal model and intervened by cordyceps sinensis. Triglyceride (TG), total cholesterol (TC) in liver were measured by biochemistry, adenosine triphosphate (ATP) by luciferin-luciferinase, and UCP2 expression by Northern blotting and immunohistochemistry. Liver histopathology was evaluated. [Results] High-fat diet fed rats developed obesity and showed a gradual increase in body weight, liver index, and a decrease in ATP level. More advanced liver disease was found histopathologically for longer high-fat diet. Up-regulation of liver UCP2 by high-fat diet stopped after week 12. However cordyceps sinensis induced UCP2 up-regulation continuously, and kept liver ATP a relatively high level. [Conclusion] NAFLD rat models were produced successfully. Liver UCP2 up-regulation in NAFLD rats may be a definite beneficial adaptation to lipid exposure. Cordyceps sinensis may serve a protective role to prevent NAFLD from progression. One of the possible mechanisms involves in modulating UCP2 expression and thereby, regulating fat metabolism, energy homeostasis.

**Key words** — nonalcoholic fatty liver, Chinese herbal, uncoupling protein 2

## INTRODUCTION

Nonalcoholic fatty liver disease has been an increasing focus of medical attention which is reported to be the most common liver disease in the world. It represents a spectrum of liver lesions that occur in individuals who either do not consume any alcohol or only consume alcohol in quantities generally considered not to be harmful to liver (usually < 20 g/day). Chronic hepatitis C and hepatic steatosis are common entities that have the potential to interact synergistically and result in significant morbidity<sup>1)</sup> (Patel K., Zekry A. *et al.*, 2005). For example, it is reported that hepatic steatosis is present in about 50% of the subjects with hepatitis c virus (HCV) and the genotype 3 infection contributes to the development

of steatosis<sup>2)</sup> (Nomura H., Kashiwagi S. *et al.*, 2004). Our nonalcoholic fatty liver diseases (NAFLD) rat model was produced just by using high fat diet without HCV infection (The concomitant model of NAFLD with HCV infection is the next step of our study).

Though mechanisms of fatty liver disease are poorly defined, there are evidences to support the popular “two-hit” hypothesis for pathogenesis. The first hit is fat accumulation in liver, which is the primary cause of fatty liver disease. Uncoupling protein 2 (UCP2), a newly identified mitochondrial inner membrane carrier protein, uncouples fuel oxidation from adenosine triphosphate (ATP) synthesis<sup>3)</sup> (Fleury C., Neverova M. *et al.*, 1997). It has been reported that UCP2 regulates lipid metabolism to prevent liver from steatosis<sup>4)</sup> (Cortez P. H., Zhi L. H. *et al.*, 1999). But on the other hand, it may cause hepatocyte injury for possibility of ATP decrease<sup>5)</sup> (Chavin K. D., Yang S. *et al.*, 1999). It is suggested that UCP2 plays an important role in the development of nonalcoholic fatty liver.

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*Cordyceps sinensis* (CS), called Dong-Chong-Xia-Cao (winter-worm and summer-plant) in Mandarin, is a member of the ascomycetes class of fungi that parasitizes larvae of Lepidoptera. It attacks the caterpillars in late autumn, and by early summer of the following year, has killed the caterpillar, and the fruiting body protrudes from its head<sup>6)</sup> (Zhu J. S., Halpern G. M. *et al.*, 1998). According to many ancient descriptions of Chinese herbs, CS possesses important pharmacological activities and has been used as an herbal medicine for centuries in Asian societies. As traditional Chinese medicine, CS was used to treat weakness after sickness. It has been proven to enhance hepatic function<sup>7)</sup> (Manabe N., Azuma Y. *et al.*, 2000), decrease plasma cholesterol levels<sup>8)</sup> (Koh J. H., Kim J. M. *et al.*, 2002), ameliorate liver diseases in cases of liver cirrhosis by experimental and clinical evidences<sup>9)</sup> (Zhang R. H., Zhou Z. C. *et al.*, 1999), but we lack documents of its effect on fatty liver diseases. It gives us clues to investigate its possible molecular mechanisms by observation the effect on nonalcoholic fatty liver disease and on the UCP2 expression in the high-fat diet induced rat models.

## MATERIALS AND METHODS

**Animals** — 90 Sprague-Dawley male rats ( $181.42 \pm 12.05$  g) were purchased from animal center of the Third Military University of China, and fed a normal laboratory diet for one week to stabilize the metabolic conditions. Rats were exposed to 12-hr light/12-hr dark cycle and maintained at a constant temperature of 22°C with free access to food and water. 10 rats were killed as normal group (N group). The rest 80 rats were divided into two groups randomly, model group (M group) and CS group respectively. 40 rats of M group were only given high-fat diet containing 88% normal diet, 10% animal oil and 2% cholesterol, and administered with saline by intra-gastric feeding (1.5 ml/100 g BW). The other 40 rats (CS group) were given high-fat diet, and intervened with *cordyceps sinensis* (33%, v/v, 1.5 ml/100 g BW) by intra-gastric feeding. 10 rats in each of the groups were randomly killed at week 4, 8, 12, 16 and 20 (named CS4, CS8, CS12, CS16, CS20 in CS group, and M4, M8, M12, M16, M20 in M group respectively). All animals were anesthetized at about 10:00 a.m. after 6-hr fasting, by intra-peritoneal injection of pentobarbital sodium (0.08 g/g BW). Livers were isolated immediately,

weighed, and stored under -70°C for further analysis.

**Biochemical Analysis** — Liver triglyceride (TG), total cholesterol (TC) were measured under the guide of reagents kits.

**Evaluation of Hepatic ATP Stores** — Frozen aliquots of liver about 0.4 g were homogenized on ice in 10 times volume of 0.4 mol/l HClO<sub>4</sub>, and then centrifuged for 15 min with 4000 revolutions per min under 4°C. 2 ml supernate was moved into test tube and the pH was adjusted to 7.75 with 5 mol/l K<sub>2</sub>CO<sub>3</sub>. Supernate was diluted with 3 times volume of 0.02 mol/l tris-acetic acid buffer after centrifugation. ATP concentration was measured by luminometer using commercial kits (luciferin-luciferinase was obtained from CALBIOCHEM of Germany and standard ATP sample was obtained from Sigma, U.S.A.).

**Immunohistochemistry**: Liver UCP2 protein was evaluated by immunohistochemistry. Paraffin sections were deparaffinized in xylene and dehydrated in alcohol. After treatment with 3%(v/v) hydrogen peroxide in methanol for quenching the endogenous peroxidase. The samples were incubated overnight at 4°C with a 1 : 100 dilution of polyclonal UCP2 [phosphate buffered saline (PBS) as negative control] after antigen retrieval using microwave-induced epitope retrieval method. After incubation with the avidin-biotin complex, the antibody labeling using diaminobenzidine as chromogen was performed with haematoxylin counter staining and then covered with neutral gum. All sections were analyzed by Biomedical automatic analyzing system (CM-2000B BeiHang China).

**RNA Isolation and Northern Blot Analysis** — Total RNA was purified from snap-frozen liver tissue using a commercially available kit (Sigma). Reverse transcription-polymerase chain reaction (RT-PCR) was performed by using a commercial kit (Takara, Japan). After RNA concentrations and quality were confirmed. UCP2mRNA expression from each rat was evaluated by Northern blot analysis. RNA (20 µg/lane) was fractioned by electrophoresis on 1% agarose-formaldehyde gel under non-denaturing conditions, botted onto nylon membrane (Bio-Rad) by capillary transfer, and hybridized overnight at 42°C UCP2 cDNA in a solution containing 7% sodium dodecylsulfate (SDS), 50% formamide, 5 × standard saline citrate (SSC), 2% blocking reagent, 50 mmol/l sodium phosphate pH 7.0 and 0.1% SDS. After staining the membranes with methylene blue to demonstrate lane-lane variations in 18s RNA,

**Table 1.** Changes of Body Weights, Liver Indices, Lipid Content and ATP Level in All Subgroups

| Group             |                  | BW (g)               | liver indices (%)  | liver TG (mmol/l)  | liver TC (umol/l)  | ATP (umol/g)       |
|-------------------|------------------|----------------------|--------------------|--------------------|--------------------|--------------------|
| N group (n = 10)  |                  | 181.75 ± 14.12       | 3.36 ± 0.26        | 2.32 ± 0.52        | 6.25 ± 0.36        | 5.74 ± 0.82        |
| M group (n = 10)  | M <sub>4</sub>   | 225.34 ± 9.79*       | 4.62 ± 0.32*       | 3.85 ± 0.36*       | 8.63 ± 0.42*       | 4.35 ± 0.56*       |
|                   | M <sub>8</sub>   | 239.29 ± 8.06*,†     | 5.08 ± 0.26*,†     | 3.98 ± 0.47*,†     | 9.06 ± 0.53*,†     | 3.87 ± 0.43*,†     |
|                   | M <sub>12</sub>  | 257.53 ± 10.80*,†    | 5.21 ± 0.43*,†     | 4.21 ± 0.53*       | 10.74 ± 0.49*,†    | 3.26 ± 0.52*,†     |
|                   | M <sub>16</sub>  | 270.64 ± 7.87*,†     | 5.73 ± 0.26*,†     | 4.89 ± 0.49*,†     | 11.17 ± 0.50*,†    | 2.69 ± 0.47*,†     |
|                   | M <sub>20</sub>  | 288.15 ± 11.51*,†    | 6.36 ± 0.50*,†     | 5.67 ± 0.51*,†     | 12.28 ± 0.56*,†    | 2.38 ± 0.54*,†     |
| CS group (n = 10) | CS <sub>4</sub>  | 194.25 ± 8.35*,**    | 3.67 ± 0.08*,**    | 2.60 ± 0.43*,**    | 6.73 ± 0.47*,**    | 4.53 ± 0.42*,**    |
|                   | CS <sub>8</sub>  | 213.03 ± 8.59*,**,*† | 3.78 ± 0.12*,**,*† | 2.68 ± 0.54*,**,*† | 7.01 ± 0.44*,**,*† | 4.39 ± 0.50*,**,*† |
|                   | CS <sub>12</sub> | 222.34 ± 9.01*,**,*† | 3.82 ± 0.05*,**,*† | 2.75 ± 0.48*,**,*† | 7.52 ± 0.48*,**,*† | 4.04 ± 0.55*,**,*† |
|                   | CS <sub>16</sub> | 231.16 ± 7.64*,**,*† | 4.01 ± 0.06*,**,*† | 2.92 ± 0.39*,**,*† | 7.98 ± 0.51*,**,*† | 3.95 ± 0.49*,**,*† |
|                   | CS <sub>20</sub> | 243.37 ± 9.88*,**,*† | 4.21 ± 0.15*,**,*† | 3.01 ± 0.52*,**,*† | 8.54 ± 0.55*,**,*† | 3.88 ± 0.51*,**,*† |

Data are means ± S.D. \* $p < 0.01$  vs. N group, \*\* $p < 0.01$  vs. the corresponding M group of the same time points, † $p < 0.01$  vs. the sub-group of previous time-point in the same group.

blots were washed at 55°C under stringent conditions, examined by PhosphorImager (Bio-Rad).

**Histopathologic Evaluation** — Formaldehyde-fixed, paraffin-embedded sections of liver were stained with hematoxylin-eosin and evaluated for the degree of steatosis, inflammation, and fibrosis. Two pathologists, blinded to treatment of animals, evaluated each liver tissue specimen based on the criteria proposed by Brunt<sup>10)</sup> (Elizabeth M., Brunt M. D. *et al.*, 1999).

**Statistical Analysis** — Statistical analysis was performed with the statistical package for the social science (SPSS) 11.5 program. Results were expressed as means ± S.D. Comparison between the groups was tested using analysis of variance [one way analysis of variance (ANOVA)] and unpaired Student's *t*-test. Correlation between variables was tested by Nonparametric Rank-Sum test (Npar1 way). Statistical significance is defined as  $p < 0.01$ .

## RESULTS

### Body Weights and Liver Indices Changes

Phenotypic comparison of rats from all subgroups is summarized in Table 1. We monitored the body weights and liver indices (liver weight/body weight) since the experiment started. M<sub>4</sub> group rats showed large pale and fatty livers, 24% increase in mean body weight and 36% mean liver index compared with N group. A significant difference is shown ( $p < 0.01$ ). M group rats showed a gradual increase in body weight gain and liver indices from week 4 to 20. A 28% higher body weight and 37% higher

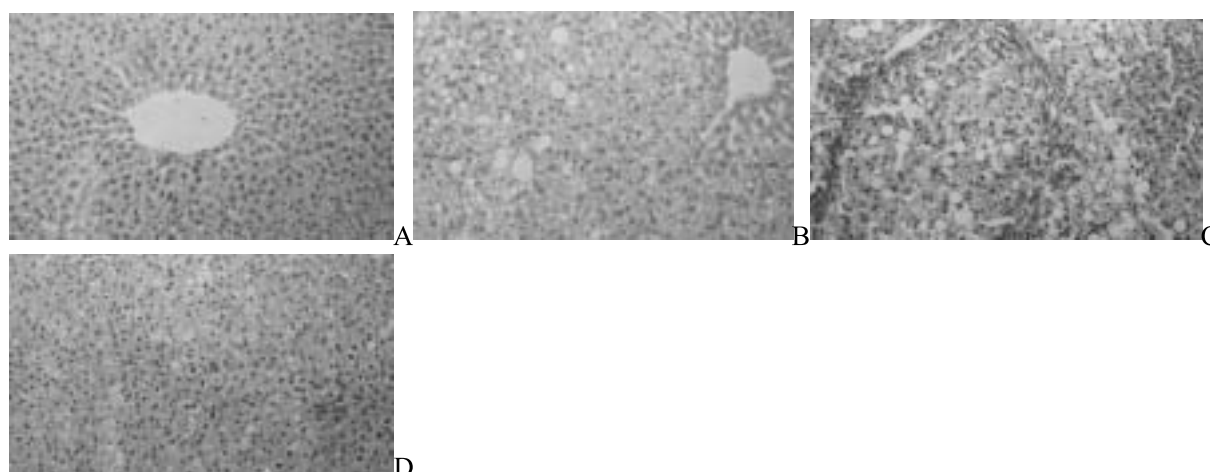
liver index was shown in M<sub>20</sub> group than in M<sub>4</sub> group. Furthermore, M<sub>20</sub> group rats weighed 1.58 time and the liver index was twice as much as those of M<sub>4</sub> group. However, both body weights and liver indices of CS group on each time-point are markedly lower than those of corresponding M group ( $p < 0.01$ ).

### Liver Histological Evaluation

Histological evaluation showed intact liver lobules without steatosis, necrosis, inflammation nor fibrogenesis in normal rats. But large and small fat droplets with normal hepatocyte configuration, were engorged in livers of M<sub>4</sub> group rats, which is defined steatosis<sup>11)</sup> (Yanai and Morimoto, 2003). However more severe steatosis, necrotic inflammation and fibrogenesis were found in livers of rats from M<sub>4</sub> to M<sub>20</sub> group. And diffusive fat infiltration, inflammatory cells infiltration, necrosis and focal and patchy even bridging fibrogenesis were found in livers of M<sub>20</sub> group, which is non-steatohepatitis even cirrhosis<sup>11)</sup> (Yanai and Morimoto, 2003). The rats of CS group exhibited diffusive fat infiltration but no markedly necrotic inflammation nor fibrogenesis in livers (shown Fig. 1 and Table 2), but which is slighter than M group rats on same time points.

### Effect of Cordyceps Sinensis on Fat Metabolism

Markedly higher levels of TG and TC were observed in livers of rats in CS<sub>4</sub> group than in normal rats ( $p < 0.01$ ). Moreover, both TG and TC increased gradually from M<sub>4</sub> to M<sub>20</sub> group. Till week 20, 1.47 time of TG and 1.42 time of TC was shown in M<sub>20</sub> group as much as in M<sub>4</sub> group. Compared with



**Fig. 1.** Representative Photomicrographs of Livers

Hematoxylin-eosin stains show no steatosis in N group (A), severe macrovesicular steatosis predominantly in zone 3 but neither inflammation nor fibrosis in M<sub>4</sub> group (B). Progressive pathogenesis, more severe steatosis, necrosis and fibrosis is detected from M<sub>4</sub> to M<sub>20</sub> group (C). Diffusive steatosis but not markedly necro-inflammation nor fibrosis is shown from CS<sub>4</sub> to CS<sub>20</sub> group (D) (original magnification 100 ×).

**Table 2.** Histological Evaluations Histological Degree (Cases)

| Group    |                  | mild | moderate | severe |
|----------|------------------|------|----------|--------|
| N group  | (n = 10)         | 0    | 0        | 0      |
| M group  | M <sub>4</sub>   | 8    | 0        | 0      |
| (n = 10) | M <sub>8</sub>   | 6    | 2        | 0      |
|          | M <sub>12</sub>  | 3    | 3        | 1      |
|          | M <sub>16</sub>  | 3    | 2        | 3      |
|          | M <sub>20</sub>  | 1    | 2        | 5      |
| CS group | CS <sub>4</sub>  | 8    | 0        | 0      |
| (n = 10) | CS <sub>8</sub>  | 8    | 0        | 0      |
|          | CS <sub>12</sub> | 6    | 2        | 0      |
|          | CS <sub>16</sub> | 6    | 1        | 1      |
|          | CS <sub>20</sub> | 6    | 1        | 1      |

M group rats on same time points, a significant lower level of TG and TC was detected in CS group. there is an significant difference ( $p < 0.01$ ), but almost no different TG or TC concentration was found between the sub-groups in CS group.

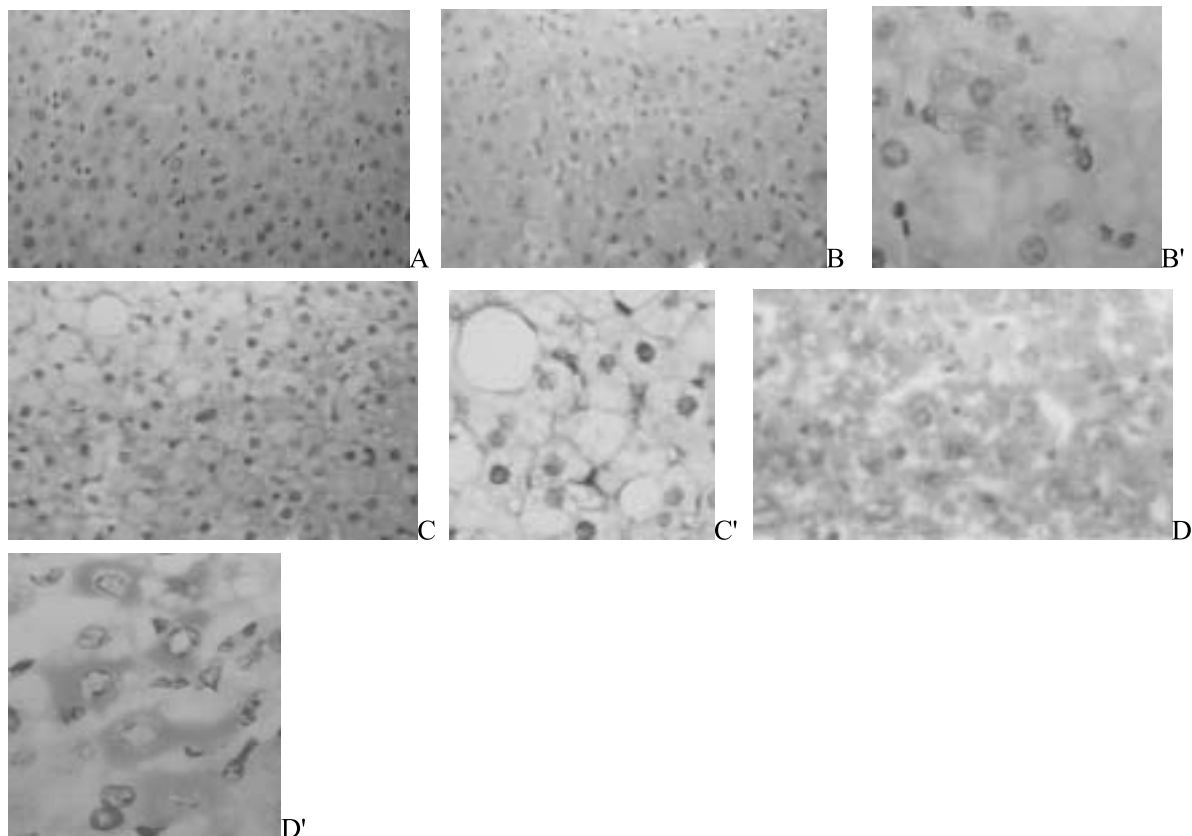
### Liver UCP2 Protein and mRNA Expression

Immunohistochemistry showed only some Kupffer cells but few hepatocytes were positive staining in livers of normal rats. However many hepatocytes as well as Kupffer cells were positively stained in livers of rats from M<sub>4</sub> group, and more and more positive hepatocytes were detected from week 4 to 12, but almost no increase from week 12 to 20 in M group. Lots of positive stained hepatocytes and Kupffer cells were also found in livers of

rats from CS group, more than those of M group on same time points. But not like M group, positive staining cell number kept going up from week 4 to 20, for cordyceps sinensis intervention (As shown in Fig. 2). Northern blot analysis showed extreme low level of UCP2 expression was detected in normal rats, but it is apparently up-regulated by high-fat diet from week 4 till 12, and almost no increase was shown from week 12 to 20 ( $p > 0.05$ ). But cordyceps sinensis induced UCP2 mRNA up-regulation continuously from week 4 to 20 under the same condition of high-fat feeding, which is similar to the change of UCP2 protein (Figs. 2 and 3).

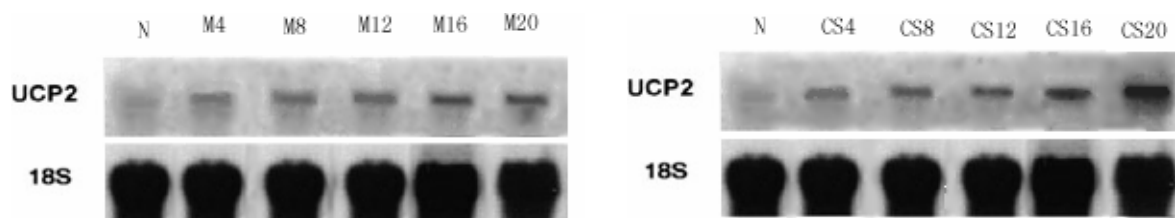
### ATP Stores in Livers

We detected ATP stores in livers with luciferin-luciferinase method. An obvious decrease in ATP contents of livers was detected in M<sub>4</sub> group rats, about 76% as much as in normal rats. Furthermore, ATP stores decreased from week 4 to 20, which is about 41% in M<sub>20</sub> group as much as in M<sub>4</sub> group. A significant lower ATP contents was found in livers of cordyceps sinensis treated rats than normal rats too, but which was markedly higher than that of M group rats on same time points ( $p < 0.01$ ), and no apparent different ATP level was found from week 12 to 20, which is summarized in Table 1 (but only UCP2 mechanism can't explain this phenomenon, it needs to be elucidated further).



**Fig. 2.** Immunohistochemical Evaluation of UCP2 Protein in Livers

UCP2 is not detected in normal livers (A  $\times 200$ ), but appears as brown staining in hepatocytes of rats in  $M_4$  group ( $M_4$ : B  $\times 200$ , B'  $\times 400$ ) and increases from  $M_4$  to  $M_{12}$ , but almost not increases from  $M_{12}$  to  $M_{20}$  group (C  $\times 200$ , C'  $\times 400$ ) which is up-regulated further by CS. It increases from CS<sub>4</sub> to CS<sub>20</sub> continuously (CS<sub>20</sub> D  $\times 200$ , D'  $\times 400$ ).



**Fig. 3.** Representative Northern Blots for Messenger RNAs of UCP2 in Liver

Total RNA (20 ug per lane) was analyzed. The corresponding blot for the 18S ribosomal RNA is also shown. Very low level of UCP2 mRNA is detected in normal rats (N). But UCP2 expression is significantly increased in the livers of  $M_4$  group ( $M_4$ ) and further increase from  $M_4$  to  $M_{12}$  group, but not almost from  $M_{12}$  to  $M_{20}$  group, which is up-regulated by cordyceps sinensis from CS<sub>4</sub> to CS<sub>20</sub> group.

## DISCUSSION

In present study, NAFLD rat models were produced by high-fat diet. From week 4 to 20, a gradual increase in body weights, liver indices, fat content, and liver pathological progression was shown in M group rats. Diffusive fat degeneration, inflammatory cells infiltration, fibrogenesis even cirrhosis were found in livers of  $M_{20}$  group, liver steatosis without

inflammation nor necrosis was kept in rats intervened with cordyceps sinensis and continuously high-fat diet feeding from week 4 to 20 (whether it develop to hepatosteatosis or cirrhosis with longer CS intervention and treatment with high-fat diet, further research is needed). The findings suggest that cordyceps sinensis can effectively restrain body weight gain and liver indices increase, and inhibit fat deposit in liver, so delay the progression from

steatosis to steatohepatitis. It gave us the impression that cordyceps sinensis take effects from multi-target points according with the multi-factor pathogenesis of fatty liver disease.

We successfully replicated rat models of non-alcoholic fatty liver disease with high-fat diet feeding, for fat deposit in hepatocytes<sup>12)</sup> (Michal C. H., Arthur I. C. *et al.*, 2004). We found triglycerides and total cholesterol raised significantly in liver of M group compared with normal rats after week 4, and both TG and TC increased gradually from M<sub>4</sub> to M<sub>20</sub> group, and a positive co-relationship was shown between high-fat diet time and liver fat loads. It verifies that lipid metabolic disturbance is not only the initiative cause of NAFLD<sup>13)</sup> (Day C. P., Saksena S. *et al.*, 2002), but it also contributes to exacerbate NAFLD, in which longer high-fat diet treatment caused more fat deposit, and so severer liver pathological change. Moreover, we found an extreme low UCP2 level in livers of normal rats which expressed prominently in kupffer cells but few in hepatocytes, however it is up-regulated increasingly in hepatocytes of NAFLD with time. It is supposed that liver UCP2 is an inducible gene<sup>5)</sup> (Chavin K. D., Yang S. *et al.*, 1999) which is a metabolic adaptation to fat disturbance, for it constrain reactive oxygen species (ROS) production and protect hepatocytes from apoptosis<sup>14)</sup> (Liang, X. F., Ogata, H. Y. *et al.*, 2003), which is also supported by the evidences that absent or insufficient UCP2 function in the regenerating liver results in increased ROS production<sup>15)</sup> (McCarty M. F., 2005) and function of UCP2 in liver is prevention of mitochondrial hydrogen peroxide generation<sup>16)</sup> (Negre-Salvayre A., Hirtz C. *et al.*, 1997). We further found that this definite response was invalidate and liver disease progressed further after 12-week high-fat diet treatment, which is not concord with the results recorded by Gu Xiao-Hong<sup>17)</sup> (Gu X. H., Zhang Y. D. *et al.*, 2002). However, under the same condition of high-fat diet feeding, livers of CS group showed continuously up-regulation of UCP2 expression, and remained simple steatosis with diffusive fat infiltration but no apparently inflammation, necrosis nor fibrogenesis. It hints that cordyceps sinensis prevents fat accumulation in hepatocytes induced by high-fat diet by up-regulating liver UCP2 expression and modulates lipid metabolism, thus plays a protective role in nonalcoholic fatty liver diseases.

Energy hemeostasis is of importance in the development of live diseases. UCP2 uncouples mitochondrial respiration from oxidative phosphoryla-

tion, so it has supposed that UCP2 may be an important regulator of energy hemeostasis of hepatocytes<sup>18)</sup> (Boss O., Hagen T. *et al.*, 2000). It was reported that up-regulation of UCP2 may reduce energy stores in hepatocytes for it reduce the efficiency of ATP synthesis, but some researches reported that UCP2 expression in liver possibly has the potential of increasing energy expenditure for prevention of obesity<sup>19)</sup> (Tsuboyama-Kasaoka N., Takahashi M. *et al.*, 1999), so we measured ATP level in livers. In present study, we found UCP2 expression is related to ATP level but not an exclusive factor to influence it. Expression of UCP2 mRNA and protein is gradually increased in M group from week 4 to 12, but almost not after week 12. More and more fat accumulation and progressive liver damage concomitant with significant decrease in liver ATP stores was shown in M group from week 4 till 20. However, Cordyceps sinensis increased UCP2 expression continuously from week 4 to 20, but not caused ATP breakdown any longer after week 12. This discrepancy between UCP2 up-regulation and ATP decrease<sup>12)</sup> (Michal C. H., Arthur I. C. *et al.*, 2004) is difficult to reconcile with the uncoupling activity of UCP2, leading to speculation that there is some co-relationship among lipids content, UCP2 expression and ATP stores in liver. Presumably, increased UCP2 induction in hepatocytes may serve a protective role by negatively modulating energy expenditure during obesity caused by high-fat diet<sup>20)</sup> (Zhang C. Y., Baffy G. *et al.*, 2001). However, extremely reduced ATP stores increased susceptibility to ATP depletion when energy needs increase acutely such as stress, thus cause advanced liver disease<sup>21)</sup> (Takahashi M., Tsuboyama K. N. *et al.*, 2002). Compared with rats treated with simple high-fat diet, rats intervened by Cordyceps Sinensis showed unceasing up-regulation of UCP2, significant higher ATP stores, and non-progressive histopathology. It illustrates that Cordyceps Sinensis can up-regulate UCP2 expression to increase thermogenesis by lipids expenditure, but on the other hand it prevents hepatic ATP depletion by some other mechanisms, so that enough energy is supplied for revival of inflamed hepatocytes.

In summary, high-fat diet fed rats developed obese and non-alcoholic fat liver disease was produced, in which induction of liver UCP2 mRNA and protein may be a definite adaptation to obesity. The present study provided evidence that Cordyceps Sinensis might be a valuable therapeutic intervention against NAFLD, the mechanisms include con-

tinuously induction of UCP2 expression and maintaining liver ATP store to a relatively high level for hepatocytes survival.

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